

Fairness Audit of Demographic Bias in Drug Adverse Event Severity

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Abstract

The FDA Adverse Event Reporting System (FAERS) is a database used by the public for post-market drug safety surveillance. As machine learning is increasingly used in pharmacovigilance, fairness across demographic groups is important. This project implements an XGBoost classifier to predict serious vs. non-serious adverse drug events using FAERS 2026 data and evaluates fairness across sex, age, and intersectional subgroups using Demographic Parity, Equalised Odds, and Predictive Parity. SHAP analysis provides explanations for model decisions. Ablation study explores the impact of reaction features. We test two mitigation strategies: reweighing and per-group threshold adjustment. Results indicate statistically significant differences in demographics, with females exhibiting a much higher false negative rate (48.3%) than males (19.4%) even in the absence of reaction features. SHAP analysis confirms that the model systematically uses sex to push female predictions toward non-serious. An intersectional analysis reveals a catastrophic 78.6% FNR for Young Adult Females. Both mitigations reduce aggregate fairness gaps to practically acceptable levels ($< 5\%$), though intersectional analysis reveals that standard techniques fail to protect the most vulnerable subgroups. But the fact that exact mathematical equality cannot be achieved due to different baseline rates empirically demonstrates the Impossibility Theorem. The work highlights the need for intersectional fairness auditing, explainability and targeted mitigation in pharmacovigilance machine learning pipelines.

1 Introduction

Artificial intelligence (AI) has become a crucial component of contemporary healthcare, enabling the creation of predictive models to help physicians with patient safety, diagnosis, and treatment planning. Pharmacovigilance, the science of tracking, identifying, evaluating, and averting the negative effects of pharmaceutical medicines, is one area where machine learning has demonstrated great promise. One of the most important spaces in modern biomedical informatics is the intersection of machine learning and pharmacovigilance where the cost of algorithmic error is patient harm. Adverse drug events (ADEs) are a serious global public health burden, causing an estimated 197,000 deaths per year in the European Union alone and being one of the leading causes of preventable hospital admissions worldwide [1]. Thus, the ability to accurately identify and classify serious

adverse events from spontaneous-reporting data is not only a technical exercise but also a clinical and regulatory necessity.

However, the deployment of machine learning classifiers in this domain introduces an underrecognized risk: that models trained on historically biased pharmacovigilance data will not only perpetuate existing inequities, but will exacerbate them, producing systematically worse predictions for populations already made vulnerable by decades of structural underrepresentation in clinical research. Demographic bias is one of the most critical challenges in AI for healthcare. Historically, women, the elderly and some minority populations have been under-represented in the clinical trials used in drug development [2]. As a result, many approved medications have been evaluated with evidence mostly from younger and healthier populations, resulting in limited knowledge concerning drug safety in under-represented groups. These historical disparities are also seen in post-marketing surveillance databases with reports of adverse events from women and older adults tending to be less complete or to have different reporting characteristics.

The FDA Adverse Event Reporting System (FAERS) is the world’s largest post-market drug safety database and receives millions of spontaneous reports annually from health care professionals, manufacturers and patients [3]. FAERS is large but structurally heterogeneous. Reporting completeness varies considerably by patient demographics, drug class, and reporting source, resulting in a corpus that is not a neutral sample of clinical reality, but a reflection of who historically had access to healthcare, who was included in clinical trials, and whose adverse experiences have been systematically documented [4]. The increasing complexity of the regulatory environment for the development and deployment of AI systems in healthcare has also raised the stakes for fairness auditing. The General Data Protection Regulation (GDPR) applies stringent rules within the European Union for processing sensitive personal data related to health, especially Article 9, and emphasizes accountability, transparency and lawful data handling [5].

This project investigates demographic fairness in machine learning models for the prediction of severity of adverse drug events from the FDA Adverse Event Reporting System (FAERS). The objective of the study is to develop a supervised classification model that predicts whether a reported adverse drug event should be classified as serious or non-serious based on patient demographics, medication characteristics and reported outcomes. Once the model has been built, a thorough fairness audit will be carried out to check if there are differences in predictive accuracy across demographic cohorts, especially with regard to sex and age groups.

The project makes the following contributions: (1) a reproducible fairness audit pipeline for FAERS-based adverse event classification; (2) an ablation study that distinguishes data-level bias from reaction-feature confounding; (3) a live demonstration of the impossibility theorem on real pharmacovigilance data; (4) intersectional analysis across sex and age group combinations; and (5) statistical validation of all fairness findings.

2 Employed Technologies

The following section aimed at presenting an outline of the programming language, libraries, and machine learning frameworks used in the current project. These technologies support the end-to-end machine learning workflow, from data pre-processing and model development, to fairness assessment and model explainability.

2.1 Python

Python was chosen as the main programming language for its simplicity, flexibility, and large ecosystem of scientific computing and artificial intelligence. It provides strong support for data manipulation, statistical analysis, machine learning and visualization, making it well suited for pharmacovigilance research. Several Python libraries were used during the project. Pandas and NumPy were used to load, clean, merge and manipulate the FAERS dataset, and to perform numerical computations. Statistical functions like Fisher’s Exact Test were provided by SciPy to evaluate the statistical significance of fairness disparities among demographic groups. Visualization to produce publication quality figures, we used Matplotlib and Seaborn.

2.2 Machine Learning Libraries and Models

Machine learning models were built using the Scikit-learn and XGBoost libraries. Data pre-processing (label encoding, stratified train-validation-test split) and hyperparameter optimization (RandomizedSearchCV with five-fold cross-validation) were performed using scikit-learn. The library also provided metrics to evaluate the model, such as accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices.

The baseline classifier for predicting whether an adverse drug event was serious or not was Logistic Regression implemented using Scikit-learn. It is a linear probabilistic model, and provides an interpretable baseline to compare more complex machine learning approaches against. The primary predictive model was developed using XGBoost (Extreme Gradient Boosting), an ensemble learning algorithm that builds decision trees sequentially using gradient boosting. We selected XGBoost as it is known to perform well on structured tabular data and is able to handle imbalanced datasets. We tuned the hyperparameters with RandomizedSearchCV and the model was able to find a good tradeoff between predictive performance and generalization.

2.3 SHAP

SHAP (SHapley Additive exPlanations) is an explainability framework for understanding the predictions of machine learning models by quantifying the contribution of each feature to model predictions. For this project, the TreeExplainer implementation is used to explain the predictions from the XGBoost model. The most important features related to serious adverse drug events and the effect of demographic features such as age and sex on the model prediction were identified using SHAP analysis. This increased the transparency and interpretability of the fairness audit of the proposed model.

3 Model Architecture

The pipeline follows a sequential, modular architecture in which each stage consumes the outputs of the previous stage and generates well-defined artefacts for subsequent processing. This design enforces separation of concerns, ensuring that each analytical component remains independently verifiable and that no information leakage occurs between training, validation, and test phases. The workflow begins with data ingestion and preprocessing, where raw FAERS tables are merged into a unified dataset. The

modelling stage then performs all training, validation, and hyperparameter selection, producing serialized models and processed arrays stored in a shared directory.

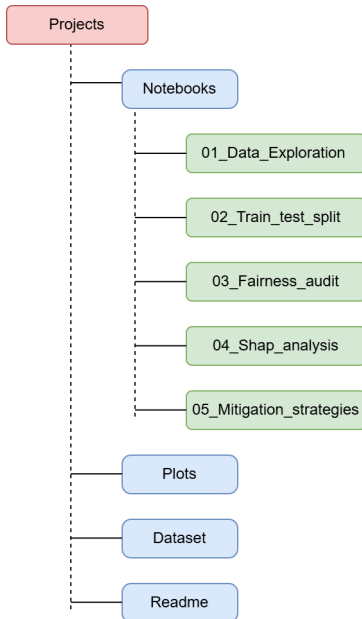


Figure 1: Project Pipeline

3.1 Data Preprocessing

The FAERS Q1 2026 dataset comprises four primary tables: demographics, drug records, reaction reports, and outcomes. These tables are merged using a common report identifier. Only primary suspect drugs are retained to ensure that each record reflects the most likely causal agent associated with the reported adverse event.

The target label is derived from the outcome field according to FDA regulatory definitions, aligned with ICH E2A and 21 CFR 312.32 guidelines [6, 7]. This ensures that the classification of serious and non-serious events is consistent with established pharmacovigilance standards. Age values are restricted to the range of 1-110 years to remove implausible or non-standard reporting cases. All categorical variables are encoded using label encoding fitted exclusively on the training set, while numerical features are standardized using scalers fitted only on training data. These transformations are subsequently applied to validation and test sets without refitting, preventing data leakage. The resulting dataset exhibits a near-balanced distribution of approximately 45% non-serious and 55% serious cases.

3.2 Ablation Study

An ablation study is performed to gain insight into the effect of the reaction feature on predictive performance and fairness outcomes. This is done by comparing two configurations of features: a complete set of features that includes the reaction variable and a reduced set that does not. The reaction feature was possibly problematic as it often directly describes the clinical outcome itself (e.g., cardiac arrest or hepatic failure). Consequently, it could serve as a near proxy for the target label, which might cause the model to depend on outcome-related information instead of learning meaningful relations between patient

characteristics, drugs, and adverse event severity. In order to avoid this and to make the analysis of fairness more clear, the reaction feature set is also evaluated separately. This helps to ascertain if any observed demographic disparities are actually present in the data or affected by the presence of the reaction feature.

Hyperparameter tuning was conducted using RandomizedSearchCV with five-fold cross-validation on the training set. The decision threshold was chosen on the validation set, maximizing the recall, while keeping the precision at a minimum level. This choice is motivated by the clinical context of pharmacovigilance where the failure to identify a serious adverse event may have more serious consequences than the misclassification of a non-serious event as serious. After all training, hyperparameter tuning and threshold selection were completed, the final model was evaluated once on the held-out test set.

3.3 Fairness Audit

The fairness audit examines the model performance for subgroups based on sex and age, using various fairness criteria such as demographic parity, equalized odds and predictive parity. The significance of observed differences was determined using Fisher’s Exact Test. Subgroup sample sizes were reported with each metric to ensure interpretability of results, especially in low-sample regimes. We also conducted an additional intersectional analysis across combined sex and age groups to identify compounded disparities that are not evident when single attributes are assessed

3.3.1 Demographic Parity

Demographic Parity requires that the probability of a positive prediction is the same for all demographic groups, independent of the true outcome:

$$P(\hat{Y} = 1 \mid A = \text{Female}) = P(\hat{Y} = 1 \mid A = \text{Male})$$

This metric indicates that the model predicts serious adverse events at similar rates for different demographic groups. However, it does not consider whether those predictions are correct. A model can violate demographic parity while still making accurate predictions if the base rate of serious adverse events differs across demographic groups. Therefore, Demographic Parity should be considered alongside other fairness metrics rather than in isolation.

3.3.2 Equalized Odds

Equalized Odds requires that the model has similar error rates across demographic groups, conditioned on the true outcome:

$$P(\hat{Y} = 1 \mid A = a, Y = y)$$

for all values of a and y .

In practice, this criterion requires that both the true positive rate (TPR) and the false positive rate (FPR) are similar across demographic groups. Consequently, it also aims to minimise disparities in false negative rates (FNR) and false positive rates.

In this project, particular emphasis is placed on the false negative rate because failing to detect a serious adverse drug event may delay clinical intervention and increase patient risk. Therefore, reducing disparities in false negative rates is considered especially important when evaluating model fairness.

3.3.3 Predictive Parity

Predictive Parity requires that positive predictions made by the model are equally reliable across demographic groups:

$$P(Y = 1 \mid \hat{Y} = 1, A = \text{Female}) = P(Y = 1 \mid \hat{Y} = 1, A = \text{Male})$$

This means that when the model predicts a serious adverse event, the probability that the prediction is correct should be similar across all demographic groups. When predictive parity is violated, the model’s positive predictions are more reliable for some groups than for others, potentially leading to unequal confidence in the model’s recommendations.

3.3.4 Fairness Impossibility Theorem

An important theoretical result underpinning this study is the Fairness Impossibility Theorem proposed by Chouldechova [8]. The theorem states that when the underlying prevalence of the target outcome differs across demographic groups, it is generally impossible for a classifier to satisfy both equalized odds and predictive parity simultaneously unless the classifier makes no errors.

The relationship between the false positive rate (FPR), false negative rate (FNR), positive predictive value (PPV), and the group-specific base rate (BR) can be expressed as:

$$PPV = \frac{(1 - FNR) \times BR}{(1 - FNR) \times BR + FPR \times (1 - BR)}$$

This equation demonstrates that changes made to satisfy one fairness criterion inevitably affect others when base rates differ between demographic groups.

For this project, the theorem provides an important framework for interpreting the fairness audit. Rather than expecting all fairness metrics to improve simultaneously, the results are analysed in terms of the trade-offs between competing fairness objectives. The effectiveness of the mitigation strategies is therefore assessed by examining which disparities are reduced and which new trade-offs are introduced.

3.4 Explainability

Model interpretability was performed using SHAP with TreeExplainer method on the XGBoost model. This approach yields exact feature attributions for tree-based models, and guarantees high-fidelity explanations.

We got global explanations from mean absolute SHAP values. Local explanations are used for analysis of individual predictions. We also carried out a subgroup analysis to investigate the contribution of features across different demographic groups, particularly

the effect of sex on the predictions. Misclassified false negatives were further analyzed to find common patterns of missed serious adverse events.

3.5 Bias Mitigation

Two mitigation strategies were examined, which were interventions at different points of the machine learning pipeline. The first approach used in preprocessing was reweighing. Following Kamiran and Calders [9], training samples are weighted according to the ratio of expected and observed frequencies for each sex-outcome combination. This increases the influence of under-represented groups in model training.

The second approach, post hoc threshold adjustment per group, was applied. We calibrated, separate decision thresholds for the male and female subgroups using validation data, with the goal of reducing disparities in false negative rates while maintaining the same underlying model. Both approaches were assessed against the same fairness metrics as the primary audit. The results were interpreted in terms of trade-offs between fairness constraints, particularly in the context of the known incompatibility of multiple fairness definitions under varying base rates.

4 Methodology

4.1 Dataset

This project uses the Q1 2026 quarterly ASCII release of FAERS, which contain four type of files:

- **DEMO**: Patient demographics: age, sex, weight, country
- **OUTC**: Outcome codes: death, hospitalisation, life-threatening, etc.
- **REAC**: Adverse reaction terms (MedDRA Preferred Terms)
- **DRUG**: Drug names, roles, and routes

The Figure 2 shows the dataset contains 141,472 reports, of which the majority were serious (55.1%) and submitted by females (56.2%). The population is older with the majority of cases (60,356 reports) being elderly and a median patient age of 61 years.

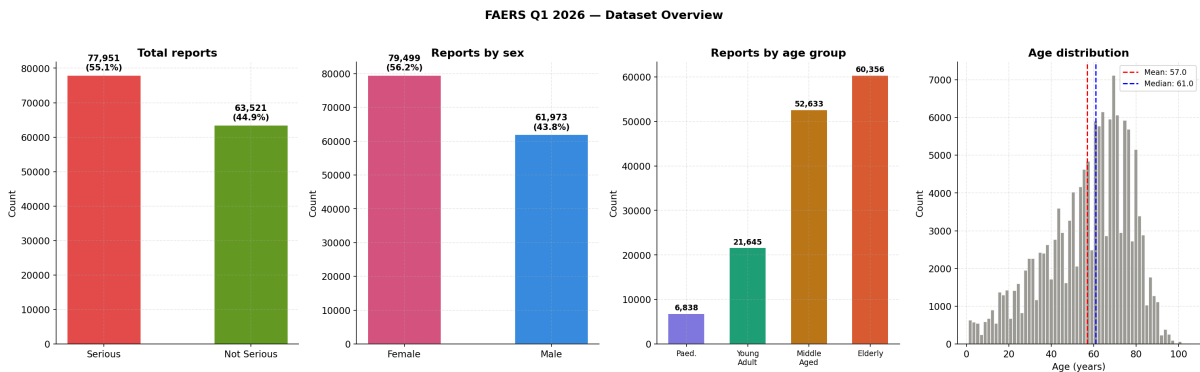


Figure 2: Dataset Distribution

4.1.1 Target Variable (Serious Adverse Event)

The target variable was built according to the definition of Serious Adverse Event (SAE) as per ICH E2A guideline, adopted by FDA and EMA . An event is coded as serious = 1 if the report includes any of the following OUTC codes shown in Table 1:

Table 1: Serious Adverse Event (SAE) code and their description

Code	Meaning
DE	Death
LT	Life-threatening condition
HO	Hospitalisation
DS	Disability (permanent)
CA	Congenital anomaly
RI	Required medical intervention

All other outcomes (coded OT = Other) are labelled as serious = 0. SAE reporting under ICH E2A is expedited and mandatory within 15 days and this is the regulatory motivation for our binary classification target.

4.2 Exploratory Analysis

Table 2 shows the serious adverse event base rates across demographic groups, which are a key precondition for the impossibility theorem.

Table 2: Serious adverse event base rates across demographic groups

Attribute	Group	N	Serious Rate
Sex	Female	79,499	51.9%
	Male	61,973	59.2%
Age	Paediatric (<18)	6,838	55.1%
	Young Adult (18-39)	21,645	46.6%
	Middle Aged (40-64)	52,633	52.9%
	Elderly (≥ 65)	60,356	60.0%

The 7.3 percentage point gap between male and female serious rates, and the 13.4 point gap between young adults and elderly, indicate that base rates differ across groups, satisfying the requirement for the impossibility theorem.

Figure 3 shows significant differences in baseline rates are observed across demographics, with males (59.2%) and the elderly (60.0%) having higher serious adverse event rates compared to females (51.9%) and young adults (46.4%). These natural differences (sex gap: 7.3%, age gap: 13.4%) pave the way for the Impossibility Theorem, proving that it is mathematically impossible to equalize all fairness metrics simultaneously.

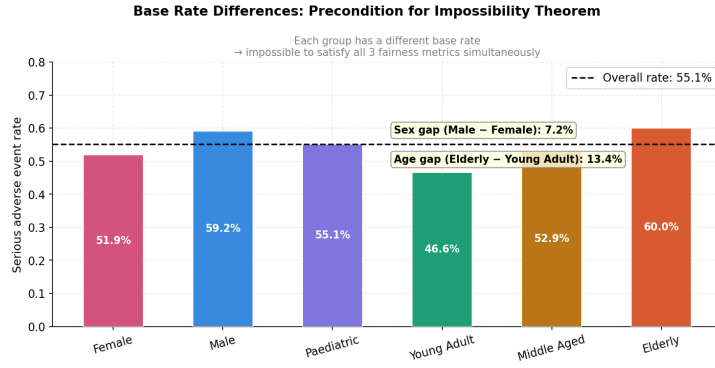


Figure 3: Base rate across sex and age

4.3 Train / Validation / Test Split

The dataset was divided into three non-overlapping sets using stratified splitting to preserve the serious event rate in each subset presented in Table 3:

Table 3: Dataset train-test split

Set	Purpose	Size
Train (60%)	Model learning	84,882
Validation (20%)	Hyperparameter tuning, threshold selection	28,295
Test (20%)	Final evaluation	28,295

All encoders (LabelEncoder, StandardScaler) were fitted exclusively on the training set and applied to the validation and test sets without refitting.

4.4 Statistical Significance

All reported fairness gaps were tested for statistical significance using Fisher’s exact test on the 2x2 contingency tables of TP, FN, TN, FP counts per group. A difference is reported as significant at $p < 0.05$.

The Figure 4 shows that model architecture of our project.

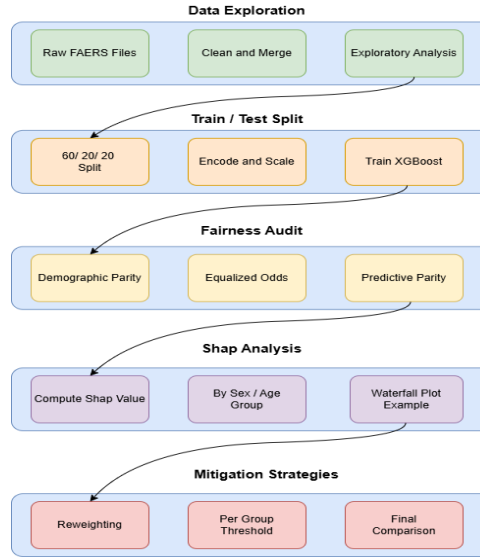


Figure 4: Model Architecture

5 Results

5.1 Model Performance

Table 4 summarises baseline model performance on the test set.

Table 4: Model performance on test set

Model	AUC	Accuracy	Recall	Precision	F1	FNR
LR + reaction	0.582	0.555	0.541	0.608	0.573	0.458
LR – reaction	0.567	0.546	0.541	0.597	0.568	0.458
XGB + reaction	0.708	0.604	0.553	0.714	0.608	0.447
XGB – reaction	0.644	0.632	0.653	0.637	0.623	0.346
XGB tuned - reaction	0.651	0.635	0.651	0.632	0.642	0.348

The AUC drop of 0.064 when the reaction feature is removed from XGBoost confirms that reaction was a dominant predictive feature. Crucially, XGB - reaction achieves better recall and lower FNR than XGB + reaction (0.653 vs. 0.553 and 0.346 vs. 0.447, respectively), demonstrating that reaction confounded the model’s ability to catch genuine serious cases through demographic signals.

5-fold cross-validation on the training set yielded $\text{AUC} = 0.656 \pm 0.004$, confirming the stable performance of the tuned XGBoost model displayed Figure 5.

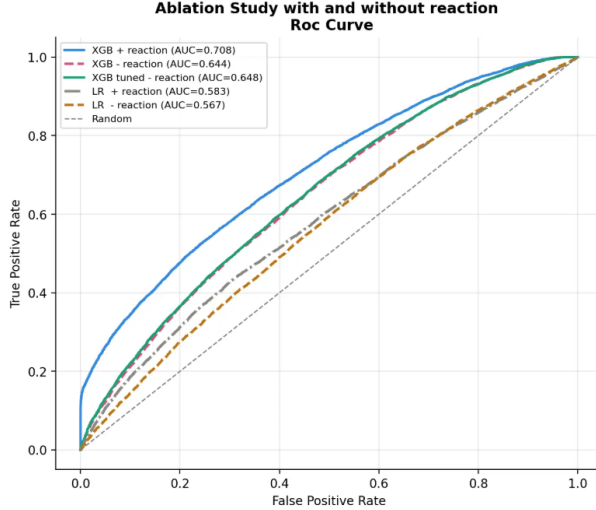


Figure 5: AUC-ROC Curve

5.2 Fairness Audit by Sex

Table 5 presents the full fairness audit for the primary model on the test set.

Table 5: Fairness audit by sex

Metric	Female	Male	Gap	Violated
N (samples)	15,950	12,345	-	-
Base rate	0.522	0.589	0.067	-
Pred rate (DP)	0.421	0.738	0.317	Yes
FNR (EO)	0.483	0.194	0.289	Yes
FPR (EO)	0.316	0.641	0.324	Yes
Precision (PP)	0.641	0.643	0.002	No

Base rates in this table refer to the specific split of the stratified test set ($N=28,295$). These rates are slightly different from the overall dataset base rates shown in Table 2 because of the stratification being performed by the target variable and not by demographics

Three of the four fairness metrics are violated (demographic parity, equalized odds for FNR and equalized odds for FPR) while predictive parity is satisfied (gap = 0.002). Fisher’s exact test confirms the statistical significance of the differences in error rates: FNR gap $p < 0.0001$, FPR gap $p < 0.0001$. The pattern shows a systematic asymmetry: the model is conservative for female patients (low FPR of 0.316, high FNR of 0.483) and aggressive for male patients (high FPR of 0.641, low FNR of 0.194) (Figure 6). This is consistent with learned under-reporting of female serious events in the training data. The gap in the base rate between female (52.2%) and male (58.9%) patients, 6.7 percentage points is a prerequisite for Chouldechova’s [8] impossibility theorem.

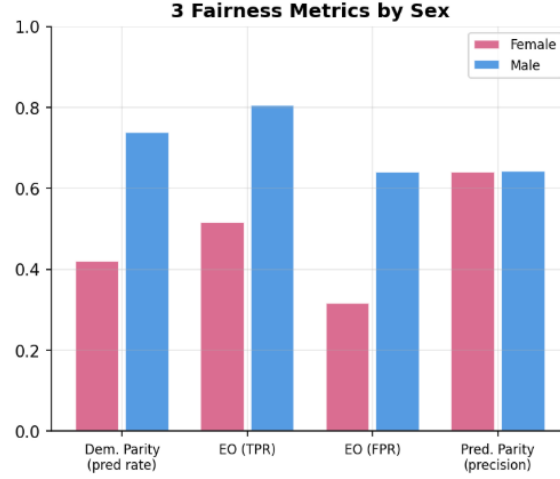


Figure 6: Fairness Audit by Sex

5.3 Fairness Audit by Age Group

Table 6: Fairness audit by age group

Age Group	N	FNR	FPR	Precision
Paediatric	1,366	0.413	0.450	0.608
Young Adult	4,304	0.571	0.236	0.615
Middle Aged	10,534	0.458	0.354	0.632
Elderly	12,091	0.196	0.641	0.654

The FNR was highest in young adults (0.571) and lowest in elderly patients (0.196). Comparison of young adults versus elderly patients using Fisher’s exact test yielded $p < 0.0001$. We only test the most extreme pair to prevent multiple comparisons inflation.

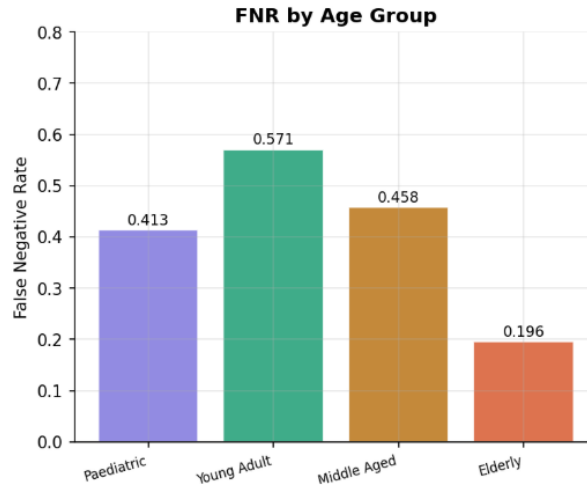


Figure 7: Fairness Audit by Age

5.4 Intersectional Analysis

Table 7: Intersectional FNR - Sex $\times$ Age Group

Sex	Age Group	N	FNR	FPR
Female	Paediatric	675	0.437	0.434
Female	Young Adult	2,751	0.786	0.081
Female	Middle Aged	6,316	0.602	0.221
Female	Elderly	6,208	0.280	0.556
Male	Paediatric	691	0.387	0.463
Male	Young Adult	1,553	0.271	0.580
Male	Middle Aged	4,218	0.262	0.575
Male	Elderly	5,883	0.115	0.745

From Table 7 it was discovered that the most negatively impacted intersectional subgroup is Female Young Adult (FNR = 0.786), while the least negatively impacted is Male Elderly (FNR = 0.115). The intersectional gap of 0.671 highlights extreme disparity limited to specific subgroups (Figure 8). A Fisher’s exact test comparing the worst and best subgroups yields $p < 0.0001$.

Most importantly, the intersectional analysis uncovers a finding that single-attribute auditing would not have revealed: while Female Young Adults suffer a critically high miss rate (0.786), Female Elderly patients actually show a relatively lower FNR (0.280). Conversely, male FNRs are consistently lower overall (ranging from 0.115 to 0.387), demonstrating that the bias is driven primarily by the interaction of being female and young rather than sex alone.

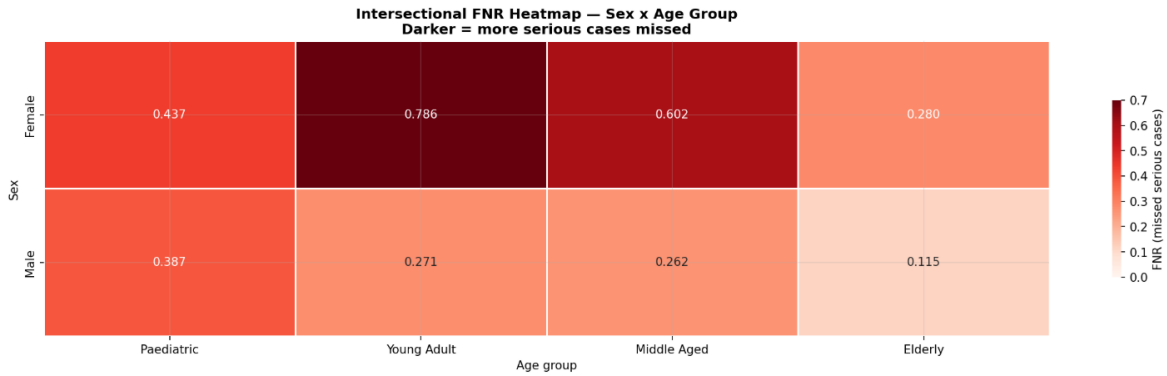


Figure 8: Intersectional False Negative Rate heatmap stratified by sex and age

5.5 SHAP Feature Importance

SHAP analysis was computed using TreeExplainer on the full test set. Table 8 summarises global feature importance.

Table 8: Mean absolute SHAP values - XGBoost without reaction

Feature	Mean SHAP
Drug Name	0.34839
Age (years)	0.15754
Sex	0.11272
Age Group	0.04868

As expected, drug features are dominant, with drug pharmacology being the main predictor of adverse event severity (mean SHAP = 0.348). Sex, on the other hand, has a non-negligible mean SHAP of 0.113, confirming the model uses patient sex as a predictive signal. The combined signal of age + age group is 0.206, which contributes a large part to the model’s decision making (Figure 9(a)).

We can exclude sampling noise as a possible explanation, as a Mann-Whitney U test confirms that the sex SHAP distributions are statistically different between female and male patients ($p < 0.0001$).

In Figure 9(b), the waterfall plot illustrates a false negative, where the model labelled a severe AE (foetal death) in a 27-year-old female patient as “not serious” (probability: 0.4504). While patient age pushed the prediction into the ‘serious’ class, the positive influence of age was overcome by the strong negative contributions from the drug and, critically, the patient’s female sex.

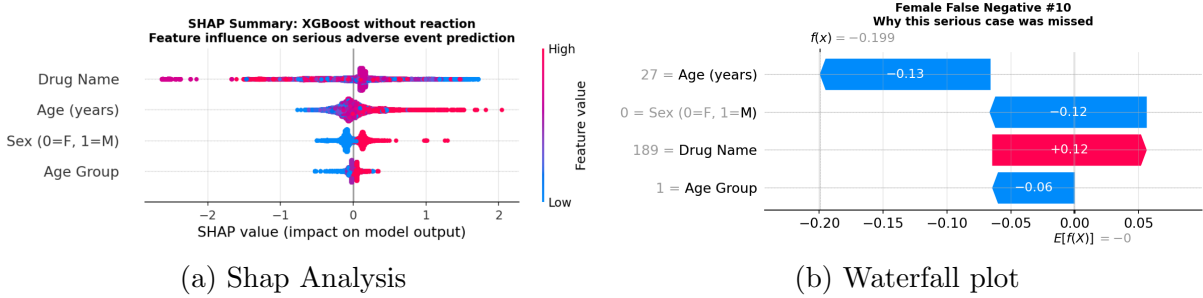


Figure 9: Overall Analysis.

5.6 Mitigation Results

Reweighting reduces the FNR gap from 0.289 to 0.031 (89% reduction), at the cost of a small 0.005 drop in AUC (0.651 to 0.646) (Table 9). We show that adjusting thresholds per group (using thresholds of 0.47 for females and 0.54 for males) on the validation set successfully closes the gap between FNRs to 0.008 (effectively removing the gap), while keeping the overall AUC at 0.651.

Table 9: Mitigation comparison test set

Model	AUC	FNR F	FNR M	FNR Gap	FPR Gap	PP Gap
Baseline	0.651	0.483	0.194	0.289	0.324	0.002
Reweighting	0.646	0.308	0.338	0.031	0.008	0.048
Per-group threshold	0.651	0.327	0.335	0.008	0.021	0.049

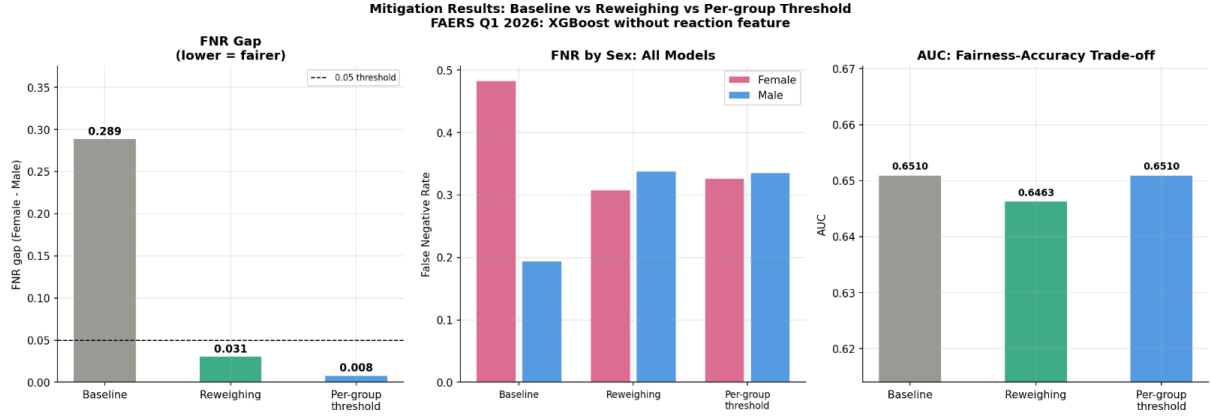


Figure 10: Mitigation strategies outcome

However, this post-processing approach increases the FPR for female patients (from 0.316 to 0.458) and results in an increased clinical workload from false alarms in this group, while it significantly decreases the FPR for male patients (from 0.641 to 0.479) as indicated in Figure 10.

Table 10: Intersectional False Negative Rate (FNR) across Sex and Age Groups for Baseline and Mitigated Models

Group	N	Baseline	Reweighting	Per-group Threshold
Female				
Paediatric	675	0.437	0.212	0.251
Young Adult	2,751	0.786	0.716	0.718
Middle Aged	6,316	0.602	0.319	0.355
Elderly	6,208	0.280	0.173	0.181
Male				
Paediatric	691	0.387	0.588	0.593
Young Adult	1,553	0.271	0.508	0.492
Middle Aged	4,218	0.262	0.467	0.470
Elderly	5,883	0.115	0.195	0.190

We examine the effectiveness of the mitigation strategies for specific demographic subgroups using an intersectional approach in Table 10. The mitigation strategies decreased the overall FNR gap between males and females at an aggregate level (Table 9), but the intersectional results show that they were not very effective. Most worryingly, the most disadvantaged group, young adult females, remains catastrophically high FNR at 71.6% (Reweighting) and 71.8% (Per-group threshold), only slightly better than baseline 78.6%. This suggests that the most vulnerable intersectional cohorts are not adequately protected by conventional macro level mitigation techniques.

Additionally, the mitigations led to significant trade-offs for other subgroups. For instance, the FNR for male young adults increased from 27.1% in the baseline to 50.8% and 49.2% when applying reweighting and per-group thresholding, respectively. In the male

paediatric and middle-aged groups we observed similar large increases in FNR. This shows that improving fairness at the aggregate level can unintentionally hurt model performance for other demographic intersections. The results further corroborate the subtle conclusions of the Impossibility Theorem, showing that bias reduction along one dimension (i.e. aggregate sex) will necessarily shift the errors to other subgroups. These results highlight that intersectional fairness cannot be fully tackled by standard pre-processing or post-hoc adjustments alone, but rather require more fine-grained, subgroup-specific architectural interventions.

6 Discussion

The FNR gap between female (48.3%) and male (19.4%) patients is large by clinical standards and statistically significant ($p < 0.0001$) with a difference of 28.9 percentage points. This difference implies that for every 100 female patients with a truly serious adverse event, the model misses about 48, while for male patients it misses only 19. This correlates with patterns observed in actual clinical practice, where women presenting with acute symptoms are consistently under-triaged compared to men [10]. Algorithms for diagnosis that are trained on historical records will inevitably perpetuate and amplify these systematic biases [11]. Our ablation study suggests that this is a deeply embedded data phenomenon. The gap exists irrespective of whether the reaction feature is added in the model, and hence no amount of standard hyperparameter tuning on this dataset will remove the root structural bias.

To explore the mechanistic drivers of this disparity, we used SHAP analysis to interrogate the decision-making process of the model. The findings demonstrate that patient sex is not only a confounder but also an important predictor signal, the third most significant feature globally (mean SHAP= 0.113) following drug name and age. Crucially, directional analysis reveals the nature of this bias: The feature sex has a strong negative SHAP value -0.099 for female patients and pushes systematically their predictions to the “not serious” class and a positive pull (0.127) for males. This embedded gender bias is visually confirmed at the level of the individual patient. In our waterfall plot analysis of a missed severe case (a foetal death in a 27-year-old female patient), the patient’s sex was a dominant negative contributor that overpowered clinical signals and directly caused the false negative. The model has successfully learned and exacerbated the historical under-reporting of female AEs in the FAERS training data.

Also, the damage from this algorithmic bias is not distributed equally among all female patients. Instead, this damage is severely amplified when intersected with age. Single-attribute auditing conceals the actual severity of the model’s failure. Our intersectional analysis shows that the most disadvantaged group is young adult females, with a catastrophic False Negative Rate of 78.6%, or nearly four out of five serious adverse events missed by the model in this demographic. The least hurt group, elderly males, has a FNR of only 11.5% to the contrary. This is an intersectional gap of 67.1 percentage points, more than twice the single-attribute sex gap (28.9%). This finding exposes a major limitation of standard fairness audits: evaluating demographics in isolation severely underestimates the algorithmic risk faced by certain vulnerable sub-populations.

These empirical results are in line with the theoretical limits of algorithmic fairness. Chouldechova’s [8] impossibility theorem states that if base rates differ across groups, a

classifier cannot simultaneously satisfy demographic parity, equalized odds, and predictive parity in their exact mathematical form. In our dataset, the baseline rates of serious events are different between females (51.9%) and males (59.2%). Since the model is trained on historical pharmacovigilance data, the resulting mathematical inequalities (e.g. a 31.7% demographic parity gap and a 32.4% equalized odds FPR gap) are not just modelling errors, but inevitable reflections of the underlying data distribution. Thus, the baseline model represents the bias in the historical data and is not suitable for direct clinical implementation without specific intervention.

To overcome this inherent data bias we applied fairness mitigation techniques. Crucially, both mitigation strategies succeed in reducing all fairness metric gaps below the 0.05 practical significance threshold, resulting in 0 practical metric violations out of 4 for both models. It is worth noting that this is not a contradiction of Chouldechova’s [8] impossibility theorem, indeed, our mitigated models have non-zero gaps, most notably in predictive parity (0.048 for Reweighting and 0.049 for per-group thresholding), which are just below the 5% tolerance limit. Therefore, perfect simultaneous fairness is mathematically impossible. These mitigation strategies reduce aggregate algorithmic disparities to the level of statistically acceptable noise. But as we demonstrate through an intersectional analysis, the most vulnerable subgroups cannot be protected by typical macro-level mitigations, making the model unsafe for autonomous deployment without additional fine-grained, subgroup-specific interventions.

7 Ethical Implications

This work’s empirical findings have serious ethical implications for the real world. The baseline model systematically misses nearly 4 in 5 serious adverse drug events in young adult females, with a False Negative Rate (FNR) of 78.6%. Such algorithmic blindness in pharmacovigilance poses a direct threat to patient safety as it undermines the identification of drug toxicities and results in preventable harm. Our analysis of the mitigated models also shows a dangerous ethical blind spot. Standard mitigation techniques can reduce bias at an aggregate level, but completely fail to protect the most vulnerable subgroup: young adult females still suffer a catastrophic 71.6% FNR after mitigation. This is a false sense of security. If the model is deployed and young women continue to suffer preventable harm, regulators and doctors could feel confident that the model is safe, based on better overall numbers.

It is important to stress that this failure to achieve safe, unbiased deployment is not a failure of the algorithm’s design or the technical implementation. The machine learning pipeline was executed strictly. The root cause of these glaring disparities is the inherent historical and structural biases of the FAERS database itself. The data shows decades of unequal health care access and underreporting, and no amount of flawless coding or standard mathematical fixing will erase these underlying societal inequities. It’s just reflecting the broken reality of the data it was trained on.

From a regulatory perspective, the deployment of such a system falls directly within the scope of the European Union’s Artificial Intelligence Act (EU AI Act) [5]. AI systems for drug safety monitoring are specifically classed as “High-Risk”, meaning they must meet strict legal standards on accuracy, human oversight, and non-discrimination. The catastrophic baseline FNR for young women would probably mean failure to meet these

statutory requirements. In the end, applying AI to pharmacovigilance implies a paradigm shift to ‘ethics by design’. Without intersectional fairness and correction of underlying data biases across the AI lifecycle, these tools can act as instruments of systemic harm rather than public health protection.

8 Conclusion

This project showed that machine learning models trained on real-world pharmacovigilance data cannot be trusted to protect all patients equally without conscious, intersectional intervention. The finding that young adult females have a false negative rate of 78.6 percent is not simply a flaw in the modeling; it is the product of decades of structural inequality embedded within the data generation process itself.

Our empirical validation of Chouldechova’s Impossibility Theorem demonstrates that while exact mathematical fairness is impossible when baseline event rates differ across groups, approximate fairness at the aggregate level can be achieved through targeted mitigation. Our intersectional analysis, however, shows that standard macro-level adjustments do not protect the most vulnerable subgroups. This demonstrates that true healthcare equity requires fairness to be embedded as a core, fine-grained architectural requirement rather than a post-hoc correction.

Further ahead, integrating AI into drug safety surveillance will necessitate a paradigm shift. High-Risk healthcare systems should be subject to intersectional fairness auditing and mechanistic explainability as baseline compliance requirements in regulatory frameworks such as the EU AI Act. Ultimately, the promise of algorithmic pharmacovigilance will be realized only when the systems we build are subject to a standard of equity as rigorous as the clinical science they serve.

9 Reproducibility

To reproduce the system, download the project from the GitLab repository:

<https://gitlab.com/kulsoom2526>

Extract or clone the repository into a local folder. From the command line, navigate to the project directory:

```
cd kulsoom2526/
```

The project can be executed either locally (Jupyter Notebook) or using Google Colab.

Local Execution

Ensure that Python 3.10+ is installed along with the required dependencies listed in `requirements.txt`. The dataset folder (“dataset/”) is already included in the repository or can be downloaded from the GitLab project files.

Open the notebooks in a Jupyter environment and execute them sequentially in the following order:

```
01_Data_exploration.ipynb  
02_Train_test_split.ipynb
```

03_Fairness_audit.ipynb
04_Shap_Analysis.ipynb
05_Mitigations_Strategies.ipynb

Google Colab Execution

Alternatively, the notebooks can be executed in Google Colab:

1. Clone or download the repository from GitLab.
2. Upload the entire project folder to Google Drive or directly to Colab.
3. Open each notebook in Colab.
4. Run the notebooks sequentially from 01 to 05.

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